

xMCF

ISO/PAS 8329

Description of mechanical connections
and joints in structural systems

ISO/PAS 8329 xMCF WG meeting
focus group on issues [#95](#) / [#104](#)
2026-07-28

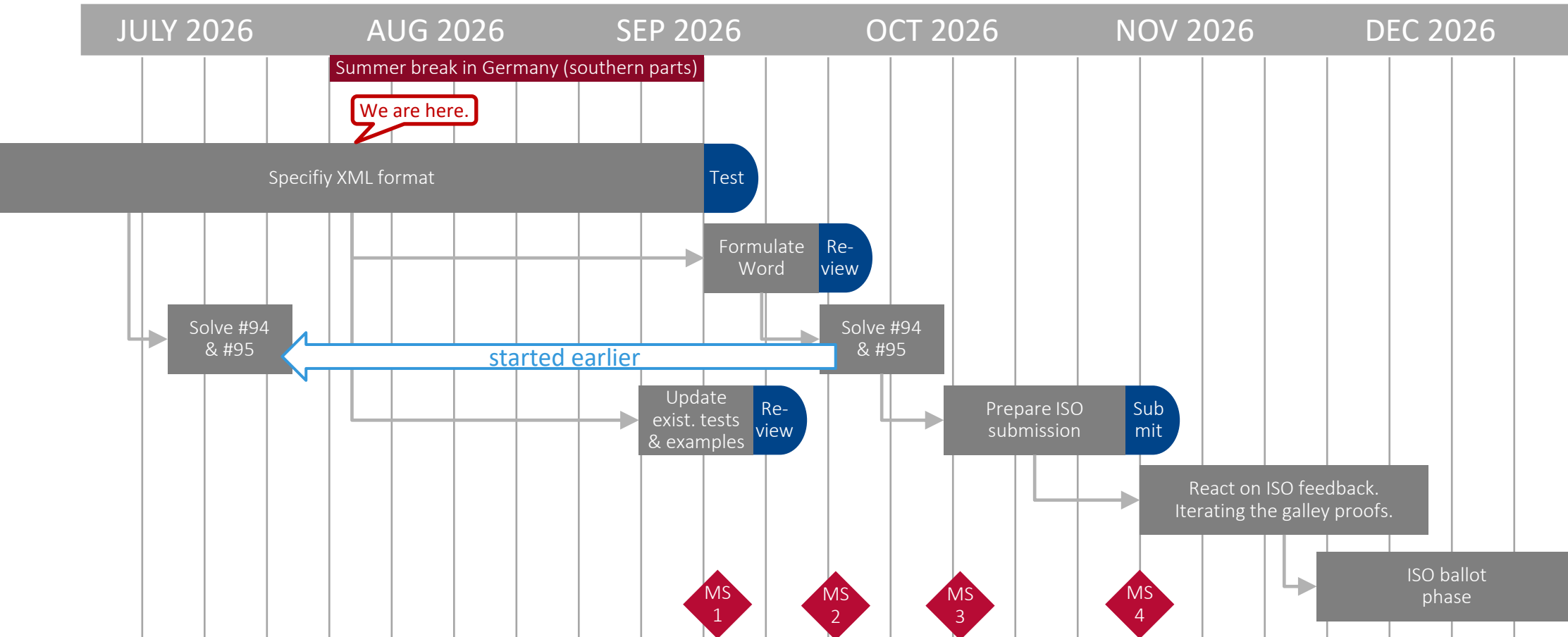
Agenda

As announced at https://github.com/economidis-nick/createXSDforxMCF/blob/master/VDA_FAT_AK_25/Meetings/2026-07-22_VideoConference/README.md

- Welcome & Flying over today's topics
- Timeline for next release
- Work in progress — focus on issues [#95](#) / [#104](#) / [#105](#) / [#61](#), as discussed last time.
- Focus on XML / DOC suggestions, XSD- / Schematron-definition & examples
- Steps we should take next



Timeline for the next release



Updating the PAS (or else striving for an IS, directly) shall be cross-checked with the chair of ISO/TC 184 / SC 4.

Issue #95: Add persistent identifier to `connection_group`

as of 2026-07-01
≙ 2026-07-08

- Original request: „Up to now, `connection_groups` have an integer identifier which is local to each `χMCF` file, and not persistent.
Under PDM considerations, an optional persistent alphanumerical id would help to increase the process reliability ...”
- This use case has been very well justified.
- However, there are other use cases of great practical relevance where the introduction of a new mandatory attribute requiring attention would result in a significant additional workload, without this being offset by any benefit.
- The group decided to **continue the discussion** at the **next meeting** with the question:
‘What if the attribute (persistent identifier) were just *optional*?’ → See next slide.
- The following important point was also raised during the discussion:
`<connected_to/>` lists the connected parts in *no particular order*.
The order may change even after a simple im- / export sequence.
If the stacking sequence is important, always use the `<stacking/>` element!



Issue #95: Add persistent identifier to `connection_group`

(continued) 'What if the attribute (persistent identifier) were just *optional*?'

Results of
2026-07-22
& 2026-07-28

- PDM Use Cases: Exchange with Partner and recognize a known connection group upon re-import.
- CAD Use Cases: Creating, Deleting, Splitting, Merging, Editing of connections, incl. changing the connected parts.
- PEP is “owner” of the unique connection group IDs, just as with the part codes.
(Single Point of Truth to guarantee uniqueness!)
PDM system, as “incarnation” of the PEP, is the tool which creates and keeps authority on such IDs.
- Difference between part codes and connection group IDs:
For CAD data, there is (usually) only *one* CAD authoring system, which is tightly integrated with PDM.
For connection data, this frequently is different:
Many tools are established for manipulating connections, which are not tightly PDM-integrated.
- Solution ideas:
 - a. PDM system offers and tool attaches to a standardized interface for querying suitable connection group ID.
 - b. Tools could exchange one xMCF file per connection group
(which renders the connection group ID as meaningless, as it is currently.)
 - c. **More ideas? --- homework!**
- Connection group ID shall be string instead of integer. — ~~This request has been split off to issue #111.~~ Rejected.
- Unfortunately, there was no time left to discuss issue [#104](#).



Issue #104: Add persistent identifier to connection_0/1/2d

On 2025-11-26, we had noted:

Difference between connection_[012]d.label and part_code?

→ See issue #104.

⇒ Maybe, formulation in standard document needs improvement.

Please leave a hint in #104, in this case.

- The issue-#94-discussion of last week (2026-07-01) fits here, as well:
 - The persistent identifier already exists. It is the connection attribute “label”.
 - The attribute “part_code” of e.g. <threaded_connection/> is applicable only for connections that are established by a physical element (then, with the given part_code).

⇒ Suggestion to close the issue as ‘wontfix’.

Issue #94: Add instance identifier to a connector

- Original request: „In order to keep consistency between the PDM system and the connection data, it is required to have an instance id for the rivets“
(Rivets or any other connection method in which a separate physical component establishes the connection.)
- The standard documents says in clause “8.2.2 Attribute `label`”: “Any connection should have an attribute called `label`, which identifies it throughout the entire CAx process, maybe even throughout the complete product lifecycle, including manufacturing. It is not necessary that these labels are unique.”

Remark 1: The use cases in which “`label`” is not unique are quite specific and rare.

Remark 2: “`label`” must not be confused with “`part_code`”, which frequently is *not* unique. Example:

```
<connection_0d label="BOLT_135" part_code="BOLT-NUT-Combi #17"> ... <!-- only label is unique -->
  <threaded_connection ... part_code="M10x50 8.8"/> <!-- not unique -->
</connection_0d>
<connection_0d label="BOLT_136" part_code="BOLT-NUT-Combi #17"> ... <!-- only label is unique -->
  <threaded_connection ... part_code="M10x50 8.8"/> <!-- not unique -->
</connection_0d>
```

- The group decided that no action is required in this case. The issue will be closed as ‘wontfix’. // ~~Done 2026-07-27.~~

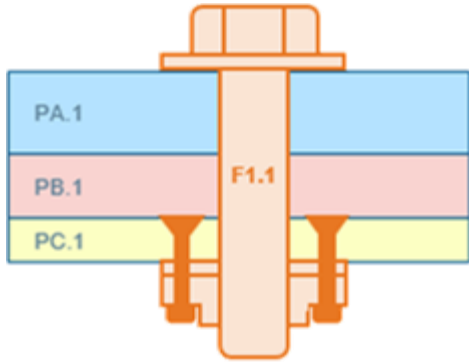
GH: Issues was closed too early: In PDM, e.g. `label="BOLT_135"` would denote an *instance* combination of e.g. a bolt, some washers and a nut. However, a part code for this combination is needed and not yet supported by χMCF.

⇒ Issue shall be re-opened. Suitable place for the `part_code` for the combination to be found in χMCF syntax – at both, the actual connection plus its template, if there is one. Suggestion: Put the `part_code` as shown above in blue, at both positions. **Homework: Please validate!**

Issue #106: Support complex rivet-nut fasteners

Due to the composition of today's attendance list, the focus was placed on issue #106.

The task is to describe a design like this:



The discussion showed that this type of modelling is feasible. It can be described using the published standard.

GH will add a note to STEP's recommended practices and also post a comment on issue #106.

The issue will then be closed. // Done 2026-07-27.

GH outlines a solution and invites discussion:

1. The nut is riveted to part PC.1.
⇒ One `connection_group` between PC.1 & nut.
2. The bolt connects PA.1, PB.1, and PC.1.
The nut is modeled "`fixed_to`".

I created this slide
after our meeting.
Is it correct, so far?

Next meetings — Suggested new appointments

- ~~2026-08-05, 15:00 CEST — focus group on [#105](#) / [#61](#) — Has been rejected 2026-07-21.~~

— *Summer break* —

- 2026-09-02, 15:00 CEST — focus group on [#105](#) / [#61](#)
— objective: agree on XML suggestions & examples
- 2026-09-09, 15:00 CEST — focus group on [#105](#) / [#61](#)
— objective: agree on XML suggestions & examples

CF sent / will send out invitations.

